JOURNAL OF GEOPHYSICAL RESEARCH, VOL. 100, NO. C3, PAGES 4489–4507, MARCH 15, 1995

A numerical study of dense water formation and transport on a shallow, sloping continental shelf

Glen Gawarkiewicz and David C. Chapman

Physical Oceanography Department, Woods Hole Oceanographic Institution, Woods Hole, Massachusetts

Abstract. The circulation and transport of dense water generated by an idealized coastal polynya is studied using a three-dimensional primitive equation model. Starting with a homogeneous, quiescent ocean, a constant negative buoyancy flux is imposed at the surface over a half-elliptical region adjacent to the coastal boundary on a gently sloping continental shelf. The flow response can be divided into the following three phases: geostrophic adjustment, instability, and offshore eddy transport. During geostrophic adjustment the fluid within the forcing region becomes denser and the flow at the edge of the forcing region accelerates in response to the strong density gradient there. Eventually, the flow at the leading edge of the forcing region (relative to Kelvin wave propagation) becomes unstable and a train of counterrotating eddies develops. These eddies then form a complex three-dimensional flow field and rapidly transport dense water offshore, across isobaths. The density within the forcing region reaches a maximum which remains fairly constant after the eddies begin to transport the dense fluid offshore. The results are qualitatively insensitive to weakening of the negative buoyancy forcing and to changing the bottom slope. Eddy scales and velocities are consistent with observations in the Arctic. The results suggest that instability processes and eddy fluxes are important in transporting dense water off continental shelves and into marginal seas.

1. Introduction

Fridtjof Nansen was one of the first oceanographers to identify the importance of continental shelf processes in the formation of dense water masses of the Arctic and North Atlantic Oceans. *Nansen* [1906] identified the eastern portion of the Barents Sea, along the western coast of Novaya Zemlya, as a preferred site of in situ dense water formation because of the presence of higher salinities than those found in the North Atlantic Water flowing into the Barents Sea from the Norwegian Sea. He concluded that the brine, rejected during ice formation on the continental shelf, led to the creation of dense water masses and that the dense water in the eastern Barents Sea might pass into the Eurasian basin of the Arctic Ocean between Franz Josef Land and the northern tip of Novaya Zemlya, where it would contribute to the abyssal waters of the basin. While more recent observations by *Swift et al.* [1983] have shown only a limited contribution of the shelf-formed water to the deepest Arctic water masses, *Aagaard et al.* [1981, 1985] have established that dense shelf waters formed during the winter are important in the maintenance of the halocline of the Arctic Ocean, although quantitative details of the shelf contribution are a matter of active debate [*Martin and Cavalieri*, 1989]. Nevertheless, numerous recent observations have identified dense water on Arctic continental shelves and related marginal seas in winter [e.g., *Schumacher et al.*, 1983; *Muench et al.* 1988; *Quadfasel et al.*, 1988; *Midttun*, 1985; *Blindheim*, 1989; *Melling*, 1993]. The dynamics of the transport of this dense fluid across Arctic continental shelves is the subject of our study.

As *Nansen* [1906] suggested, negative buoyancy forcing in polar regions is concentrated in areas of high rates of ice formation, i.e., leads and polynyas. Typically, coastal winds drive ice offshore, leaving the nearshore waters exposed to the atmosphere. As new ice is formed, brine is rejected and acts as a salt flux in the region of the polynya [*Pease*, 1987]. The divergence of the ice field associated with offshore wind forcing usually has a small offshore horizontal scale of 10 to 20 km owing to the competing effects of ice production and ice advection. (By contrast, negative buoyancy forcing in midlatitudes occurs primarily through heat fluxes, with much larger horizontal scales determined by the synoptic scale of the atmospheric forcing.) The salt flux acts as a negative buoyancy source near the coastal boundary, producing lateral density gradients in the shelf waters, which, in turn, drive a baroclinic circulation involving both the locally created dense water, as well as the ambient fluid. The presence of a sloping bot-

Copyright 1995 by the American Geophysical Union.

Paper number 94JC01742.
0148-0227/95/94JC-01742$05.00

tom over the continental shelf introduces a gravitational component which may allow cross-isobath transport of the dense water. Frictional boundary layers presumably contribute to cross-isobath transport as well, while the coastal boundary itself constrains the flow and allows the possibility of coastal currents.

Some aspects of Arctic dense water flows have been considered in previous studies, but they have all been limited by rather severe approximations which preclude many processes that we expect (and find) to be important. For example, *Kowalik and Matthews* [1983] used a two-dimensional model to study flows of dense water from Arctic lagoons, but they ignored the effects of rotation which are important on the space and timescales of polynyas. *Melling and Lewis* [1982] applied *Smith*'s [1975] stream-tube model to the flow of dense water from the Beaufort Sea into the Canadian Basin of the Arctic Ocean. However, stream-tube models assume a flow which does not allow baroclinic instability, unsteady effects, or cross-stream variations. H.-M. Hsu (The response of the shallow polar shelf water to negative buoyancy forcing: A numerical study, submitted to *Journal of Physical Oceanography*, 1993 hereinafter referred to as submitted manuscript) used a two-dimensional primitive equation model to study negative buoyancy forcing adjacent to a coast and found that the dense water did not move offshore beyond the forcing region unless an along-shelf current was added. However, since there was no along-shelf variation in the flow field, neither plumes nor eddies could be formed in the model.

In this study we use a numerical model, the semispectral primitive equation model (SPEM) of *Haidvogel et al.* [1991], to study the three-dimensional, time-dependent response of an initially unstratified fluid to negative buoyancy forcing within a limited region. The primary questions to be addressed concern the nature of the physical processes involved in the initial stages of offshore transport of dense water. Is transport primarily via a quasi-steady, nearly geostrophic current analogous to a stream tube, or are instabilities and eddy-induced transport processes dominant? What role does the bottom boundary layer play in the motion of the dense fluid? Does the sloping bottom enhance offshore transport? We are primarily interested in studying particular fundamental dynamical processes, rather than simulating an oceanic flow complete with tides, winds, ice, and realistic bathymetry, so we choose rather idealized geometry, forcing, and parameterizations. Our hope is that the insight and basic understanding gained from this work will apply to more complex situations and will stimulate further modeling and observational efforts which may include the interplay of wind and buoyancy forcing in the presence of ice effects. The paper is organized as follows. The numerical model, SPEM, is described in section 2. An analytical two-layer model [*Dewar and Killworth*, 1990] which is useful in anticipating some of the features of the three-dimensional flows is briefly described in section 3. One example of the response to constant negative buoyancy forcing over a limited area is examined in detail in section 4. The model sensitivity to variations in several of the parameters is described in section 5, and the results are discussed in section 6. Finally, a summary is given in section 7.

2. Model Description

We use a semispectral primitive equation ocean circulation model, described in detail by *Haidvogel et al.* [1991], to solve the following hydrostatic and Boussinesq momentum, density, and continuity equations. (*Jones and Marshall* [1993] have recently shown that nonhydrostatic effects are important in the development of deep convective plumes. They also examined one case of convection in relatively shallow water and found the nonhydrostatic effects to be unimportant. The present calculations fall in this shallow water regime.)

$$u_t+(uu_x+vu_y+wu_z)-fv = -\frac{1}{\rho_0}p_x+(A_V u_z)_z+F_u \quad (1)$$

$$v_t+(uv_x+vv_y+wv_z)+fu = -\frac{1}{\rho_0}p_y+(A_V v_z)_z+F_v \quad (2)$$

$$p_z = -\rho g \quad (3)$$

$$\rho_t + u\rho_x + v\rho_y + w\rho_z = (K_V \rho_z)_z \quad (4)$$

$$u_x + v_y + w_z = 0 \quad (5)$$

In this system (u, v, w) represent the (x, y, z) components of the velocity with z directed vertically upward from the ocean surface; ρ is the difference between the total density and a constant reference density ρ_0; p is the pressure; f is the Coriolis parameter; g is gravitational acceleration; A_V is the vertical eddy viscosity; K_V is the vertical eddy diffusivity; and t is time. The variables $F_{u,v}$ represent dissipative functions which are required for numerical stability. Subscripts x, y, z, and t denote partial differentiation.

The system (1)–(5) is approximated using finite differences in the horizontal, a stretched vertical (sigma) coordinate to handle variations in the bottom topography, and a high-order spectral approximation (an expansion in modified Chebyshev polynomials) to represent the vertical flow structure. A leapfrog-trapezoidal time-stepping scheme is used with an occasional trapezoidal step correction. *Haidvogel et al.* [1991] give further model details.

The model is configured in a high-latitude, uniformly rotating ($f = 1.3 \times 10^{-4}\mathrm{s}^{-1}$), straight channel (Figure 1) with a coastal wall of depth h_0 at $y = 0$ and an offshore (solid) wall at $y = 75$ km. The channel is 150 km long with periodic boundaries at the ends. The channel is wide enough and long enough, that the boundaries do not significantly influence the evolution of the dense water during the time of interest here. The bottom deepens linearly away from the coast with slope s;

$$h(y) = h_0 + sy \quad (6)$$

The coastal wall depth is taken as $h_0 = 50$ m, while the standard bottom slope is $s = 0.001$. There is no flow through the bottom, and a rigid lid is assumed at

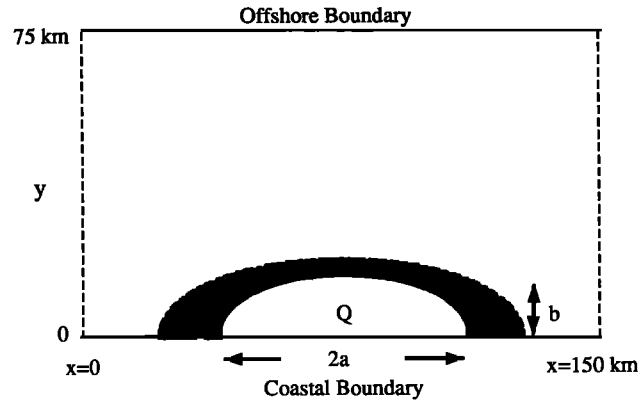

Figure 1. A plan view of the computational domain, a periodic channel 150 km long and 75 km wide. The offshore ($y = 75$ km) and coastal boundaries ($y = 0$) are solid, while periodic boundary conditions are applied at $x = 0$ and $x = 150$ km. The negative buoyancy forcing is limited to a half-elliptical region adjacent to the coastal boundary. Shading indicates the forcing decay region.

the surface (i.e., $w = 0$ at $z = 0$). The numerical grid is uniform in the horizontal with 145 along-channel and 73 across-channel grid points, i.e., $\Delta x = \Delta y = 1.03$ km. Tests with twice the resolution showed virtually identical results. Nine Chebyshev polynomials are used to resolve the vertical structure (see the appendix). The model time step is 288 s. At the surface the stress is set to zero,

$$A_V u_z = A_V v_z = 0 \qquad z = 0 \tag{7}$$

The surface density flux ($z = 0$) is specified by

$$\begin{aligned} K_V \rho_z &= Q & r(x,y) < 1 \\ &= \left(\frac{Q}{2}\right)\{1 + \cos[2\pi(r-1)]\} & 1 < r(x,y) < 1.5 \\ &= 0 & r(x,y) > 1.5 \end{aligned} \tag{8}$$

where $r^2(x,y) = (x - 75\ \text{km})^2/a^2 + y^2/b^2$, and Q is a constant density flux (Figure 1) which represents the effect of brine rejection in a coastal polynya, a is a half width in the along-shelf direction which is 30 km for the standard case, and b is a width in the cross-shelf direction which is 10 km. These horizontal scales for wind-driven coastal polynyas are based on the observations of *Schumacher et al.* [1983] in the northern Bering Sea and the simple model of *Pease* [1987] which balances ice production with ice advection. The region in which the forcing decreases ($1 < r < 1.5$) is henceforth referred to as the "forcing decay region." This idealized form of the density flux obviously ignores wind effects and the details of the actual freezing process. However, it allows us to examine the effects of a negative buoyancy forcing in isolation, which is the focus of our study. At the bottom the shear stress is specified using a linear bottom friction parameterization,

$$A_V u_z = ru \qquad z = -h \tag{9}$$

$$A_V v_z = rv \qquad z = -h \tag{10}$$

where the bottom friction coefficient is set to $r = 0.0005$ m s^{-1}. There is no density flux through the bottom, which becomes

$$K_V \rho_z = 0 \qquad z = -h \tag{11}$$

because of the neglect of lateral mixing of density in (4).

The vertical mixing coefficients are approximated using a Richardson number dependent scheme based on *James* [1977] and *Munk and Anderson* [1948]:

$$A_V = 0.0001 + \frac{0.0049}{(1. + 0.3Ri)^{1/2}} \tag{12}$$

$$K_V = 0.0001 + \frac{0.0049}{(1. + 0.3Ri)^{3/2}} \tag{13}$$

where the units are square meters per second and the Richardson number is defined by

$$Ri = \frac{-g\rho_z/\rho_0}{(u_z)^2 + (v_z)^2} \tag{14}$$

Additional vertical mixing of density is applied in the form of instantaneous vertical convective adjustment whenever the water column becomes statically unstable (i.e., when lighter water appears under heavier water). Convective adjustment is implemented when needed beneath the forcing region, but the standard explicit vertical diffusion in (1), (2), and (4) dominates outside the forcing region. We would prefer to eliminate explicit lateral mixing entirely, however, some lateral mixing of momentum is necessary to obtain numerically stable solutions. Therefore we apply Laplacian subgrid-scale mixing with a constant mixing coefficient along sigma coordinate surfaces (i.e., $F_{u,v} = \nu_u \nabla_\sigma^2 u, v$) and use the smallest mixing coefficient which produces a stable calculation, $\nu_u = 50$ m^2 s^{-1}. There is no lateral subgrid-scale mixing of density, except in the case of a steep bottom slope (section 5) in which Laplacian mixing with a coefficient of $\nu_\rho = 5$ m^{2}s^{-1} is used. (Note that this does not alter the bottom density flux boundary condition because the lateral mixing is applied along sigma surfaces.) The free-slip (no-stress) boundary condition is applied at the sidewalls, as are no-flux conditions for density and mass.

Each calculation begins from rest. The fluid in the channel is initially homogeneous with $\rho = 0$. At time $t = 0$ the surface density flux is applied and held fixed for 30 days. The standard density flux is $Q = 2.5\times10^{-5}$ kg m^{-2}s^{-1} which is based on the limited estimates from observations. This is smaller than fluxes observed during individual events [e.g., *Schumacher et al.*, 1982] but represents a similar total flux when averaged over many events. The sensitivity of the results to a weaker flux is discussed in section 6.

3. Expectations

Before describing the complex three-dimensional flow resulting from the numerical computations, it is worthwhile to consider two simpler situations which will help us understand the more complicated flows.

Immediately after the surface density flux Q is imposed, the density field will mix vertically from surface to bottom, but lateral advective processes will not have begun. We expect that the density anomaly within the forcing region will initially simply increase with time, roughly as $\rho \approx Qt/H$, where H is the average depth within the forcing region. That is, a front will form at the edge of the forcing region, and the density difference between the fluid within the forcing region and the ambient fluid outside the forcing region ($\rho = 0$) will be a linearly increasing function of time.

As rotation effects become important (within an inertial period), the density front will begin to adjust geostrophically. Geostrophic adjustment of a two-layer fluid has been considered over a flat bottom in which a vertical interface initially separates fluids with a fixed density difference $\Delta\rho$ [*Dewar and Killworth*, 1990, Appendix A]. Dewar and Killworth showed that the final shape of the density interface (i.e., the thickness of the denser fluid layer) is given by

$$h_2(y) = (H/2)[1 + \cosh(y - y_t) - \frac{1+\cosh(D)}{\sinh(D)} \sinh(y - y_t)] \tag{15}$$

where $y_t < y < y_b$, H is the water depth, and y is nondimensionalized by the baroclinic deformation radius

$$r_D = (g\frac{\Delta\rho}{\rho_0}\frac{H}{2})^{1/2}/f \tag{16}$$

The positions y_t and y_b are the locations of the surface and bottom density outcrops, respectively, after the adjustment. The distance $y_b - y_t = D$, where D is a constant which is given by $D \tanh (D/2) = 2$ and is roughly equal to 2.4. The thickness of the light fluid is $h_1 = H - h_2$. The velocities within the two layers are in opposite directions and are given by

$$u_1 = (fr_D/2)\{\sinh(y_b - y) - \frac{1+\cosh(D)}{\sinh(D)} [\cosh(y_b - y) - 1] - (y_b - y)\} \tag{17}$$

and

$$u_2 = (fr_D/2)\{\sinh(y - y_t) - \frac{1+\cosh(D)}{\sinh(D)} [\cosh(y - y_t) - 1] - (y - y_t)\} \tag{18}$$

The velocities reach a maximum of

$$|v_{\max}| = (fr_D/2)[\sinh(D) - \frac{1+\cosh(D)}{\sinh(D)} - D] = 1.2(fr_D) \tag{19}$$

at the surface and bottom outcrops. The shape of the frontal interface and the velocities in the two layers are shown in Figure 2.

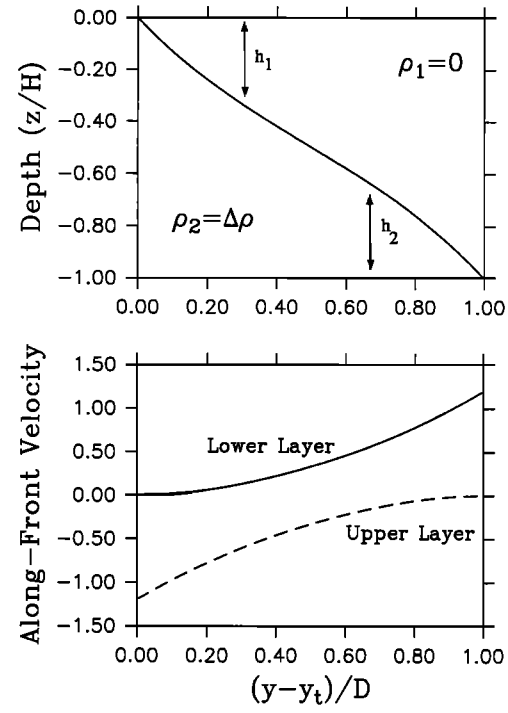

Figure 2. (top) The density profile and (bottom) velocities for an inviscid two-layer fluid after geostrophic adjustment [from *Dewar and Killworth*, 1990, Appendix A]. The velocities are scaled by fr_D from (16).

On the basis of the above two results we may anticipate several qualitative aspects of the three-dimensional, time-dependent calculations described in the next section. First, the density difference at the edge of the forcing region will be a function of time, initially changing like Qt/H until lateral advective effects become important. If the density difference increases slowly, we expect the initial phase of geostrophic adjustment to resemble the two-layer flow, except that the flow within the frontal region will accelerate with time and the velocity fields will vary smoothly in both the vertical and horizontal. However, the presence of bottom friction breaks the symmetry of the two-layer flow, so we expect larger velocities to develop near the surface outcrop region of the front and smaller velocities near the bottom. Further, owing to the shape of the forcing region, we anticipate that the area where the forcing region intersects the coastal boundary will be dynamically important because the baroclinic flow associated with the front will be directed into or away from the wall, thus requiring a dramatic ageostrophic adjustment near the coastal boundary. As we shall see, it is this region, where the front intersects the coast, which dominates the dynamics of the resulting flows.

4. Constant Negative Buoyancy Flux Over a Limited Spatial Region

We examine the Eulerian and Lagrangian flows associated with a constant negative buoyancy flux of $Q = 2.5 \times 10^{-5}$ kg m^{-2} s^{-1} imposed over a limited spatial

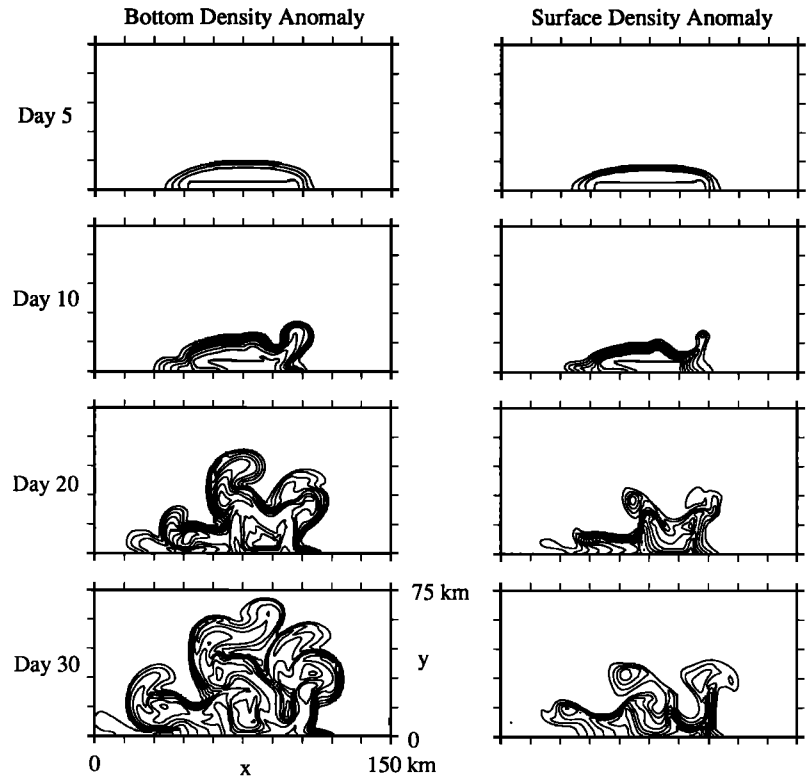

Figure 3. Plan views of the density anomaly at the (left) bottom and (right) surface after 5, 10, 20, and 30 days. The contour interval is 0.05 kg m^{-3}, the most offshore contour being 0.05 kg m^{-3}.

extent on a gently sloping continental shelf ($s = 0.001$). The forcing region is a half-elliptical area adjacent to the coastal boundary (Figure 1) with a cross-shelf scale of $b = 10$ km and an along-shelf scale of $a = 30$ km, both scales suggested by the observations of *Schumacher et al.* [1982] in the northern Bering Sea and by the coastal polynya model of *Pease* [1987]. Both scales are large compared with the baroclinic Rossby radius, which we will see reaches a maximum of roughly 4 km. The critical question concerns the processes by which dense fluid is transported away from the forcing region. Does the flow form a quasi-steady, nearly geostrophic stream tube, or do instabilities and eddy transport processes dominate?

Plan views of the density field at the bottom (Figure 3, left) and surface (Figure 3, right) show that the flow remains confined to the forcing region for at least 5 days, but then it becomes unstable, forming eddies which transport dense water offshore. A detailed examination of the flow fields suggests that the temporal development can loosely be divided into the following three phases: geostrophic adjustment (days 0–6), instability and eddy development (days 6–10), and offshore eddy transport (beyond day 10), each of which is described next.

4.1 Geostrophic Adjustment

The geostrophic adjustment phase, anticipated in section 3, is the time during which the density beneath the forcing region increases with time and the flow begins to accelerate toward geostrophy. Figure 4 shows that the maximum density of the fluid beneath the forcing region increases almost exactly as Qt/H for about 12 days. The cross-shelf structure of the velocity fields and density field in the middle of the channel ($x = 75$ km) is shown in Figure 5. Beneath the forcing decay region ($y = 10$ to 15 km), the along-shelf velocity u is consistent with the expectations from the two-layer model of *Dewar and Killworth* [1990] (section 3), a surface-trapped flow in the $-x$ direction and a near-bottom flow in the $+x$ direction, although here with a maximum above the bottom owing to the bottom stress.

The width of the frontal region for this case does not scale with r_D as in the geostrophic adjustment of two layers, but rather more nearly scales with the width of the forcing decay region L_w, which is 5 km at $x = 75$ km. The maximum geostrophic velocity near the surface in the along-shelf direction can be estimated by assuming a thermal wind balance, so that

$$fu_z = -\frac{1}{\rho_0}p_{yz} = \frac{g}{\rho_0}\rho_y \tag{20}$$

Taking

$$\rho_y \approx \frac{Qt}{HL_w} \tag{21}$$

then

$$u_z \approx \frac{u_{\text{surface}} - u_{\text{bottom}}}{H} \approx \frac{g}{f\rho_0}\frac{Qt}{HL_w} \tag{22}$$

If we assume that $u_{\text{surface}} \approx -u_{\text{bottom}}$, then we can

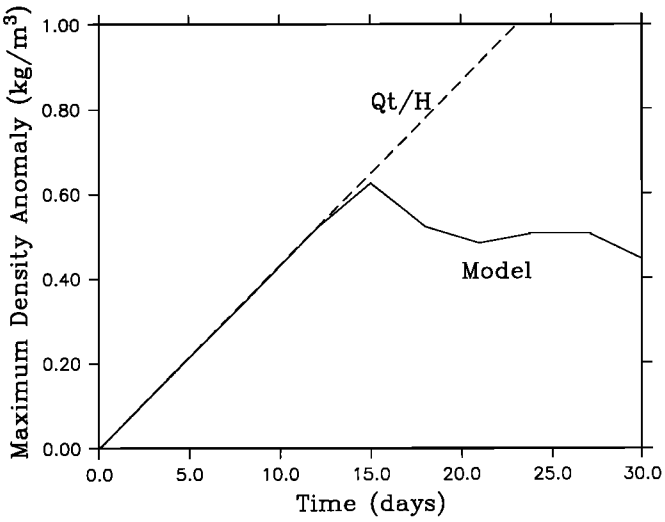

Figure 4. The maximum density anomaly within the computational domain versus time (solid curve) for the case of $Q = 2.5 \times 10^{-5}$ kg m^{-2} s^{-1} and $s = 0.001$. The linear estimate $\rho(t) = Qt/H$ (dashed curve) is shown for comparison.

approximate u_{surface} by

$$u_{\text{surface}} \approx \frac{gQt}{2f\rho_0 L_w} \tag{23}$$

Figure 6 (top) shows a comparison of the maximum along-shelf u velocities in both the $+x$ and $-x$ directions from the numerical model and the estimate (23). The magnitude of the model along-shelf velocity at the surface is slightly greater than the estimate until day 9. Near the bottom the model along-shelf velocity is fairly close to the estimate until day 4, after which it remains at roughly 0.05 m s^{-1} and is about half the estimate by day 8 owing to the influence of bottom friction.

The secondary circulation within the frontal region is vigorous. The cross-shelf velocity v consists of offshore flow in the bottom boundary layer at the base of the front with a compensating onshore flow near the surface (Figure 5). If one assumes classical Ekman dynamics with a no-slip boundary, then the maximum transverse velocity is roughly 1/3 of the geostrophic along-shelf velocity [*Pedlosky*, 1987, p. 190]. Figure 6 (middle) shows the comparison between the maximum

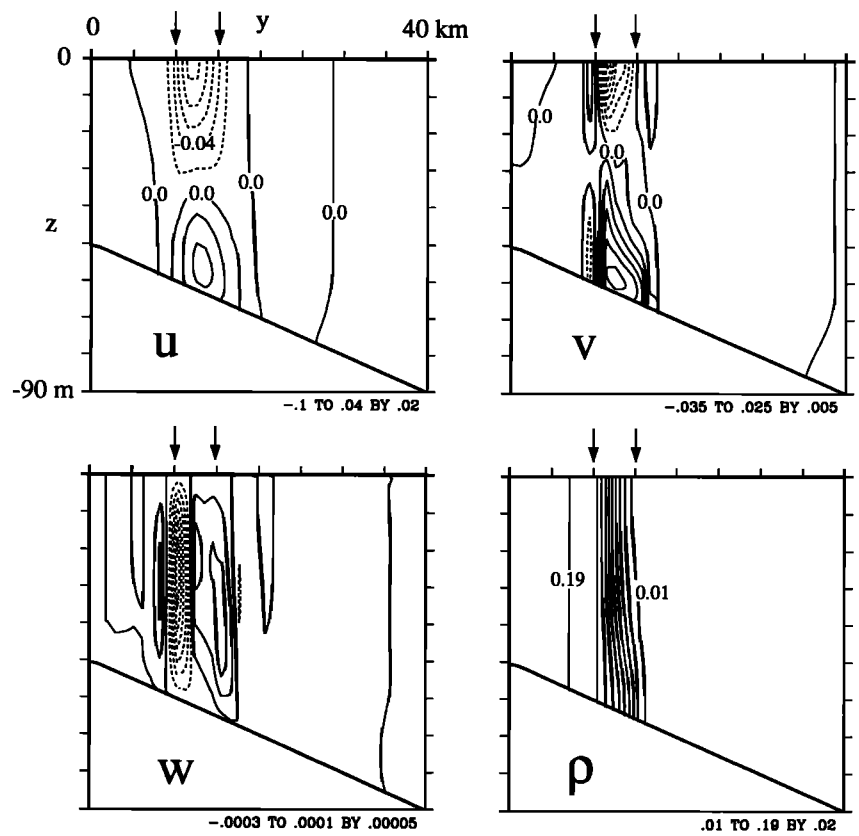

Figure 5. Cross-shelf structure of the velocity fields and density field for the standard case at $x = 75$ km, the center of the forcing region, at $t = 5$ days. Only the shoreward half of the channel is shown ($y = 0$ to 40 km). The contour intervals are 0.02 m s^{-1} for the along-shelf velocity u, 0.005 m s^{-1} for the cross-shelf velocity v, 5×10^{-5} m s^{-1} for the vertical velocity w, and 0.02 kg m^{-3} for the density anomaly ρ. Solid contours denote positive values, while dashed contours denote negative values. The inner and outer edges of the forcing decay region are indicated by the arrows, and contour values are given below each panel.

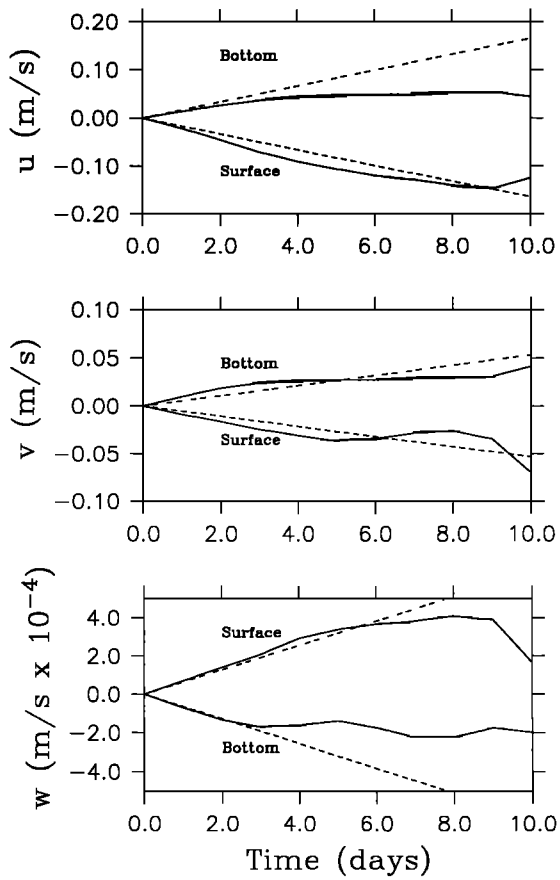

Figure 6. Maximum (top) along-shelf, (middle) cross-shelf, and (bottom) vertical velocities near the surface and bottom at $x = 75$ km based on the numerical calculations (solid curves) and geostrophic estimates (dashed curves).

model cross-shelf velocities and the estimate based on 1/3 of the geostrophic velocity (23). The scaling estimates are close to the computational values for the first 4 days.

The vertical velocity field consists of upward motion near $y = 15$ km and downward motion near $y = 10$ km, coincident with the convergence of the bottom boundary layer flow at the outer edge of the forcing region and the divergence at the inner edge (Figure 5) with the horizontal scale of L_w. An estimate for the vertical velocity can be made by assuming that $u_x = 0$ across the shelf at $x = 75$ km, so that the continuity equation becomes $v_y = -w_z$. Then using $v \approx u_{\text{surface}}/3$ and L_w and H as the horizontal and vertical scales, respectively, then $w \approx -u_{\text{surface}}H/3L_w$. Figure 6 (bottom) shows the comparison with the model vertical velocities at middepth. Again, the agreement is fairly good for the first 6 days near the surface and about 3 days near the bottom.

The density field (Figure 5) shows some slumping of the isopycnals offshore at the bottom, consistent with the two-layer model, but the structure is strongly affected by the frictional processes and is not antisymmetric about middepth as in the inviscid two-layer case. Perhaps the most important point regarding the secondary circulation and the density structure is that the dense fluid near the bottom is not transported offshore of the edge of the forcing region ($y = 15$ km) either by the adjustment process or by the Ekman transport in the bottom boundary layer. The offshore velocities vanish abruptly near $y = 15$ km where the vertical circulation is strong; that is, this is a region of large Ekman pumping out of the bottom boundary layer, consistent with the findings of Hsu (submitted manuscript, 1993) for the strictly two-dimensional case.

4.2. Instability and Eddy Development

The second phase of the flow development, instability and eddy development, begins after about 6 days, when the geostrophic flow associated with the density front starts to interact strongly with the coastal wall. Clearly, the nearly geostrophic along-front flow u shown in Figure 5 cannot continue unchanged where the front intersects the coastal boundary because the flow would be directed into or out of the boundary. Dramatic ageostrophic adjustments take place there which produce a strong perturbation to the frontal circulation. The result is the growth of an unstable wave at the leading edge of the forcing region (near $x = 110$ km; the wave appeared in Figure 3 at finite amplitude after 10 days). (We refer to the edge of the forcing region near $x = 110$ km as the leading edge because it is in the direction of Kelvin wave propagation.) Subsequently, a train of eddies grows along the frontal boundary.

Additional insight is gained by examining the temporal development of the flow field near the leading edge of the forcing region where the instabilities initially develop. Figure 7 shows the density field and velocity vectors at the bottom and surface after 4, 6, 8, and 10 days. At day 4 the surface flow within the frontal zone is in the $-x$ direction and away from the coastal boundary, while the bottom flow is primarily in the $+x$ direction and toward the coastal boundary. The surface velocities are larger than the bottom velocities (Figure 5) owing to bottom friction, so there is a net depth-averaged flow away from the boundary. This flow must be supplied by ambient shelf water from larger $+x$ locations (owing to vorticity constraints), thus forming a barotropic coastal-trapped current flowing in the $-x$ direction toward the leading edge of the forcing region. By day 6 the barotropic coastal current has displaced the front in the $-x$ direction, forming a sharp bend or "corner." This triggers the growth of a baroclinically unstable wave, which by day 8 has developed into a finite amplitude eddy with large offshore velocities along the trailing edge of the eddy. By day 10 the eddy begins to move offshore, away from the forcing region.

The surface density expression of the eddy is smaller than the bottom expression, giving the eddy a truncated cone shape, which is consistent with the release of available potential energy through baroclinic instability. The near-surface cyclonic flow within the eddy, coupled with the anticyclonic, near-bottom flow, gives the eddy a hetonlike structure [*Hogg and Stommel*, 1985].

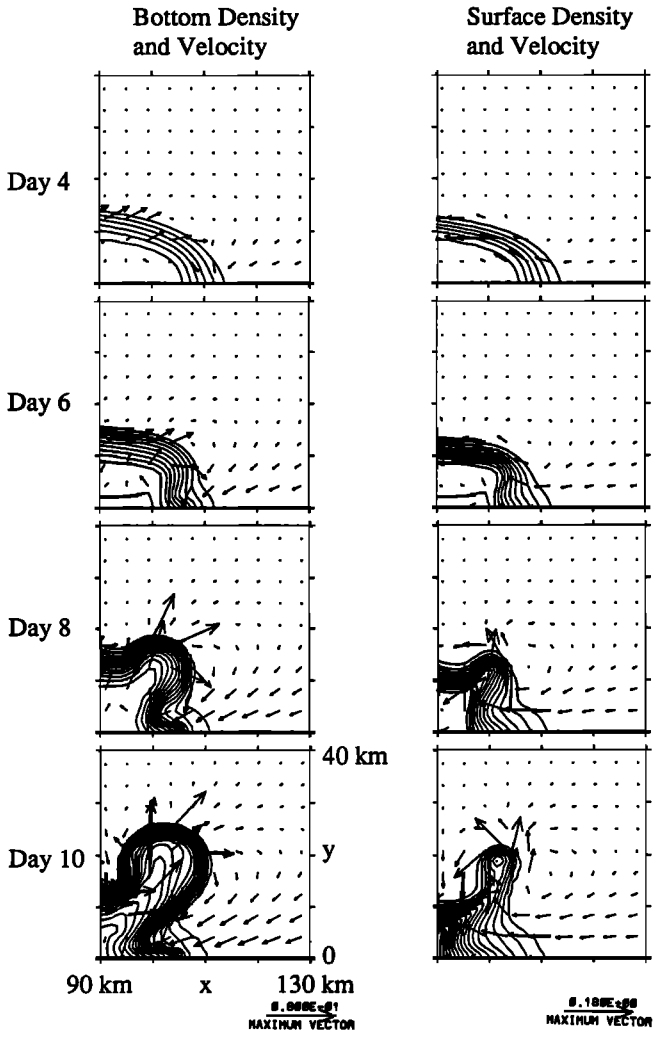

Figure 7. Plan views of the density anomaly and the velocity vectors at the (left) bottom and (right) surface near the leading edge of the forcing region at days 4, 6, 8, and 10. Vectors are drawn every fourth grid point for clarity. The maximum vectors (located in the lowest panels) are 0.08 m s^{-1} for the bottom vectors and 0.18 m s^{-1} for the surface vectors. The contour interval for the density anomalies is 0.025 kg m^{-3}.

The spatial structure of the vertical motions during the development of the eddies is complicated. Figure 8 shows the vertical velocities at middepth at days 4, 6, 8, and 10. At day 4 there is a band of downward velocities at the inner edge of the forcing decay region and a band of upward velocities at the outer edge of the forcing decay region, consistent with Figure 5. At the coast, along the leading edge of the forcing region, however, there are strong upward velocities where the surface flow moves offshore. By day 6 the downward velocities become more pronounced as the leading edge of the density front deforms and the upward velocities are stronger adjacent to the sharp density front. The eddy at the leading edge of the front develops by day 8 and contains a region of large downward velocity in the center to balance the strong radial outward flows near the bottom at the edge of the eddy owing to bottom friction (see Figure 7). The upward velocities along the coast move in the $-x$ direction but weaken. Finally, at day 10 there is a narrow band of strong ($\approx$60 m d^{-1}) downward velocities in the jet extending from the leading edge of the forcing region offshore to the eddy.

The sharpening and along-shelf asymmetry of the density front are apparent in the velocity and density fields along y = 5 km (Figure 9), i.e., an along-shelf slice. At day 1 the flow at both the leading edge and trailing edge is baroclinic, with near-bottom flow directed in the $+x$ direction. The density structure is symmetrical. By day 6, however, the acceleration of the near-surface flow results in a deepening of the flow in the $-x$ direction at the leading edge, as well as a current along the coast in the $+x$ direction within the forcing region. This current along the coast reaches a maximum velocity of roughly 0.025 m s^{-1} after 10 days which is weaker than the flow within the frontal region, where the maximum surface and bottom flows are 0.12 and 0.04 m s^{-1}, respectively (at x = 75 km). At the leading edge of the forcing region the coastal-trapped flows are in opposite directions creating a strong horizontal velocity convergence there which strengthens the density gradients dramatically by day 8. Finally, by day 10 the barotropic flow in the $-x$ direction at the leading edge of the forcing region is so strong that it advects the density front in the $-x$ direction.

An examination of the momentum balances (1) and (2) during the eddy development stage reveals that the flow near the leading edge of the forcing region away from the coastal boundary is initially geostrophic. Close

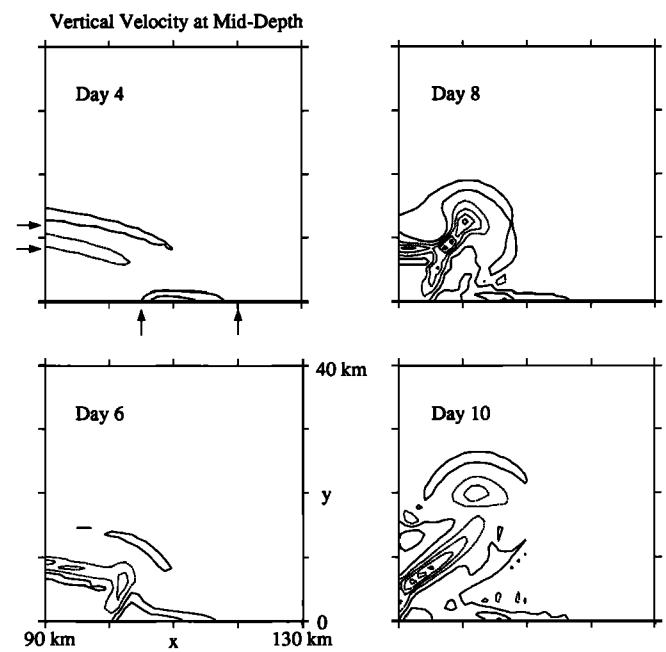

Figure 8. The vertical velocities at middepth near the leading edge of the forcing region at days 4, 6, 8, and 10. The contour interval is 0.0002 m s^{-1}. The arrows in the top left panel indicate the inner and outer edges of the forcing decay region.

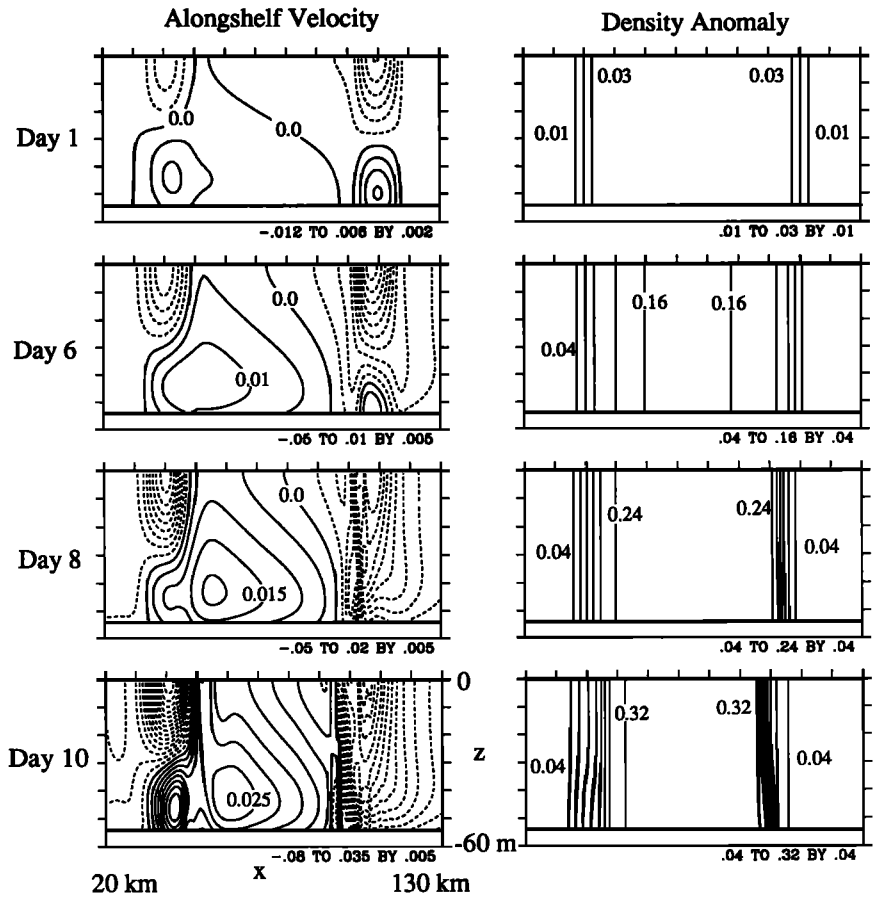

Figure 9. An along-shelf section ($y = 5$ km) of the (left) along-shelf velocity and (right) density anomaly fields at days 1, 6, 8, and 10. The contour intervals are shown below each panel.

to the coastal boundary, horizontal and vertical diffusion are comparable to the pressure gradient. However, after day 4, nonlinear advection within the frontal zone also becomes significant as the barotropic flow accelerates in the $-x$ direction. The Rossby number (ratio of the sum of the advective terms to the Coriolis term) within the frontal zone reaches 0.5 by day 4 and remains large as the eddy develops.

The horizontal scale of the first eddy which forms is roughly 15 km. This is about 5 times the baroclinic Rossby radius at $t = 6$ days and is consistent with the length scale for baroclinic instability [*Pedlosky*, 1987]. Subsequent eddies have the same scale initially before they deform while moving offshore. We have used linear instability analysis to estimate the timescale for growth of the most unstable wave, assuming no curvature of the forcing region (i.e., two dimensions with no along-front variation) and no dissipation. The model of *Duffy* [1976], which assumes a constant velocity in each layer, was applied using the density difference across the front in the numerical model at day 6 (0.26 kg m^{-3}). The velocities in each layer were taken from the maximum values in Figure 6 (0.12 m s^{-1} for the ambient layer and 0.05 m s^{-1} for the dense layer). The two-layer model predicted an e-folding time for the maximum growth rate of the instability (0.86 days) which is close to that of the numerical computations (0.75 days between days 7 and 8). The most unstable wavelength from the two-layer model was 14 km (compared with 15 km from the numerical computations), so the results from the two-layer model describe the initial phases of the eddy development quite well. The time at which the initial growth of the eddy occurs is related to the strength of the negative buoyancy forcing (see section 5.2), but it is also likely to be influenced by the magnitude of the coefficient for the lateral mixing of momentum.

4.3. Offshore Eddy Transport

The third phase of the temporal development involves the transport of dense fluid offshore by the eddies. It begins at about day 10 when the eddy at the leading edge of the forcing region begins to move offshore, away from the front. Until this time the maximum density had increased linearly with time (Figure 4). However, as eddies form and move offshore, the maximum density is limited and begins to decrease slowly after 15 days. Clearly, the eddies are carrying dense water offshore.

The flow patterns and density field along the bottom become complex as the eddies move away from the forcing region. Figure 10 shows the bottom density and barotropic stream function at days 12, 14, 16, and 18. The stream function representing the depth-averaged

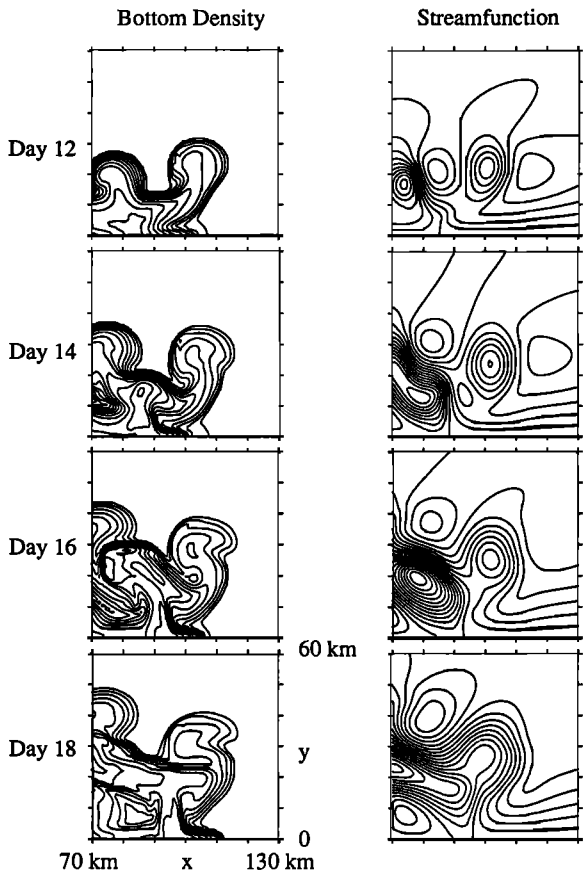

Figure 10. Plan views of the (left) bottom density anomaly and (right) barotropic stream function near the leading edge of the forcing region at days 12, 14, 16, and 18. The contour interval is 0.05 kg m^{-3} for the density anomaly and 6×10^3 m^2 s^{-1} for the stream function, with dotted contours indicating negative values.

velocity tends to indicate the flow of the near-surface fluid because the deeper flow is weakened by bottom friction. At day 12 there are two protruding features in the density field with scales of roughly 20 km. In contrast, there are four local extrema in the stream function field, and the centers of the two density features are located near the maximum gradient (i.e., maximum offshore velocity) between the extrema in the stream function field. This implies that the protruding dense water features are being advected offshore. By day 14 the patterns of both density and stream function have become more complex as the eddies have apparently interacted. The eddy at the left edge of the stream function field at day 12 has moved in the $+x$ direction and toward the coast, into shallower water, where it has strengthened, perhaps owing to vortex squashing. By days 16 and 18 the eddies have continued to interact and there are numerous filaments of dense water between the eddy centers, which are reminiscent of the laboratory experiments on eddy merging by *Nof and Simon* [1987]. The barotropic flow in the $-x$ direction adjacent to the coastal boundary is a persistent feature throughout the 30 days of the model run.

Between days 10 and 20 the center of the dense fluid at the leading edge of the forcing region moves from about $x = 102$ km, $y = 22$ km to about $x = 97$ km, $y = 31$ km, yielding mean velocities for the feature of 0.006 m s^{-1} in the $-x$ direction and 0.014 m s^{-1} in the $+y$ (offshore) direction. This predominantly offshore movement is clearly different from the path expected for isolated eddies or point vortices. For example, *Nof*'s [1983] model of an isolated, inviscid, bottom-trapped eddy on a sloping bottom produces no cross-isobath motion and would translate in the $+x$ direction. A possible explanation is that the offshore motion of the eddies in the present results is caused by self-propulsion through eddy–eddy interaction, as in the case of two point vortices which advect each other and together move along some path determined by their relative strengths. We have attempted to fit the trajectories of the eddies in the stream function field to those of the interaction of two barotropic point vortices using the method of *Zabusky and McWilliams* [1982] (with topographic beta replacing the planetary beta used in their calculations). These attempts proved unsuccessful; the point vortex trajectories were quite erratic for initial conditions similar to the velocities on day 8 of our computations. Trajectories similar to those in the numerical computations could be obtained only for vortex strengths much greater than those in our computations and with greater cross-isobath velocities. We expect that bottom friction and baroclinicity would alter the comparison significantly, but a more important influence may be interactions resulting from the continued development of eddies at the edge of the forcing region.

The details of the dense water transport by the eddies are quite complex; however, viewed together, the eddies tend to move offshore and transport dense fluid offshore throughout this time period. The integrated effect of the eddy advection can be estimated from the total cross-isobath mass flux M_V, defined as

$$M_V = \int_0^{150\ \mathrm{km}} \int_{-h}^{0} (v\rho) dz dx \qquad (24)$$

where the integration is from the bottom to the surface in the vertical and along the entire channel in the horizontal. Figure 11 shows time series of M_V at the offshore edge of the forcing region ($y = 15$ km) and roughly at the middle of the channel ($y = 35$ km). As expected, the total mass flux across $y = 15$ km increases dramatically after the development of the instabilities at around day 7 and then becomes erratic as more eddies develop and move offshore, the mean being about 16,600 kg s^{-1}. The mass flux at $y = 35$ km does not become appreciable until the eddy near the leading edge of the forcing region passes this line. The mean mass flux at $y = 35$ km is roughly 7200 kg s^{-1}, considerably less than at $y = 15$ km. By way of comparison the total mass flux into the forcing region is 18,500 kg s^{-1}, so the mean offshore flux is 90% of the surface input past $y = 15$ km but only 40% of the surface input past $y = 35$ km for the 30 days of the calculation.

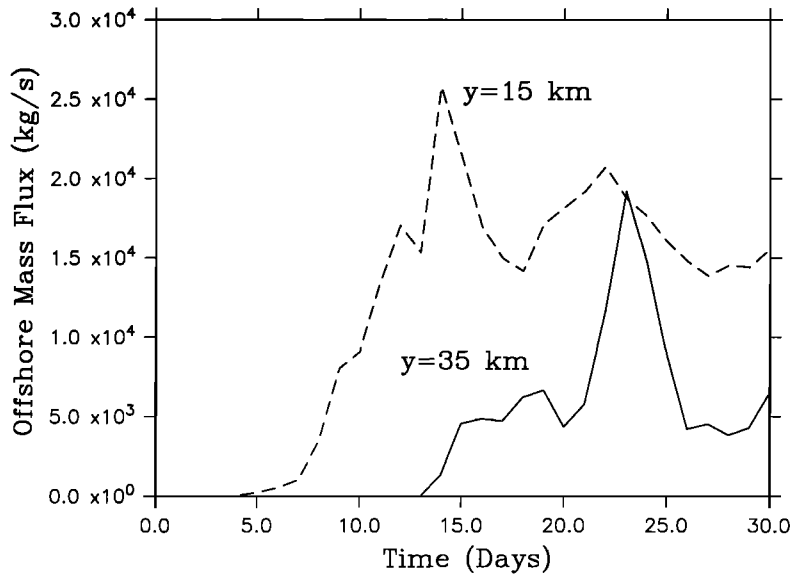

Figure 11. Time series of total offshore mass fluxes, computed from equation (24), across $y = 15$ km (solid curve) and $y = 35$ km (dashed curve) for the standard model run.

4.4. Neutrally Buoyant Particle Motions

We have described the temporal development of the Eulerian velocities in this complex, inherently three-dimensional flow. We now use neutrally buoyant particles to examine the Lagrangian nature of the flow. Particles were released 5 m below the surface and 3 m above the bottom along three separate cross-shelf lines as follows: near the trailing edge of the forcing region, in the center of the forcing region, and near the leading edge of the forcing region. Particles were tracked using a second-order Runge-Kutta scheme implemented by *Hedstrom* [1990], and locations were recorded every day for 30 days. Figures 12 and 13 show plan views and cross-shelf projections of some of the particle trajectories.

Near-surface particles released near the trailing edge of the forcing region quickly bifurcate in the area where the isopycnals intersect the coastal boundary (Figure 12a). Some of the particles are carried in the $-x$ direction and remain near the surface. However, particles released close to the coastal boundary are carried in the $+x$ direction by the circulation beneath the forcing region, move to the bottom in the downwelling zone adjacent to the coast (Figure 12b), and are then carried offshore along the bottom. The maximum offshore distance traveled by one of these particles is 35 km.

Near-surface particles released near the middle of the forcing region also separate, with particles originally near the coastal boundary downwelling and being carried up to 60 km offshore by the eddy field (Figure 12c). Particles originally near the offshore edge of the forcing region are also carried offshore in eddies originating near the middle of the forcing region, undergoing large and rapid vertical excursions from surface to bottom (Figure 12d).

At the leading edge of the forcing region the near-surface particles are carried toward the convergence zone near $x = 100$ km, where they become entrained in the eddy formed at the leading edge of the forcing region (Figure 12e). All of these particles are carried downward once they enter the eddy circulation, except for the particle released farthest offshore (Figure 12f).

The motion of the near-bottom particles is equally complicated, but the vertical excursions are not as dramatic. Near the trailing edge of the forcing region, particles tend to move in the offshore and $+x$ directions while within the bottom boundary layer, but they are rapidly carried in the $-x$ direction once they rise out of the bottom boundary layer (Figures 13a and 13b, particles B and D). The particle released closest to the coastal boundary (particle A in Figure 13a) becomes entrained in the eddy which forms near the leading edge of the forcing region, but the other particles are not carried far in the $+x$ direction. Particles released in the middle of the forcing region are also carried offshore significant distances but remain in the lower half of the water column, except near the coastal boundary (Figures 13c and 13d). Finally, the particles released near the leading edge of the forcing region are carried in the recirculation into the convergence zone at $x = 100$ km, where they upwell and then are carried offshore in the eddies from the leading edge and center of the forcing region (Figures 13e and 13f).

The particle motions illustrate the rich and complex structure of the three-dimensional flow. They show that the coastal boundary is an important area for upwelling and downwelling which allows near-surface particles to enter the bottom boundary layer and near-bottom particles to reach the surface. The near-surface cyclonic flow moves particles rapidly in the $-x$ direction when they rise out of the bottom boundary layer at the offshore edge of the forcing region. Particles which become entrained in the near-bottom flow of the eddies may be carried offshore rapidly, traveling 60 km offshore in 30 days.

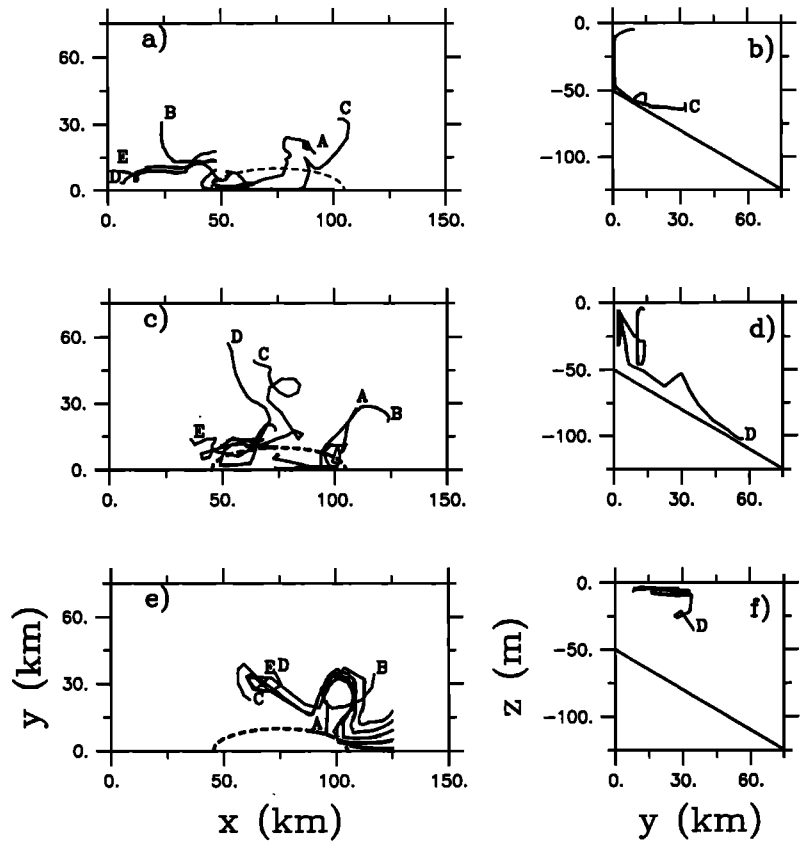

Figure 12. The 30-day trajectories of neutrally buoyant particles released 5 m below the surface: (a) plan view for particles released near trailing edge of forcing region, (b) cross-shelf projection for the same particles, (c) plan view for particles released within the center of the forcing region, (d) cross-shelf projection for the same particles, (e) plan view for particles released near the leading edge of the forcing region, (f) cross-shelf projection for the same particles. Both plan views and cross-shelf projections (of the particle moving farthest seaward) of the particle paths are shown. The trajectories are plotted at 1-day intervals. The particles are lettered such that A is initially closest to the coast and E is farthest from the coast.

5. Sensitivity to Forcing and Parameters

The results presented in section 4 are robust with variations in the amplitude of the surface forcing, the shape of the forcing region, the bottom slope, and several other parameters. That is, the same three phases of the flow response are evident and the offshore transport of dense water is associated with eddies. For the cases considered here (i.e., uniformly sloping bottom with no along-shelf topographic variations) we have not seen the development of what might even loosely be considered a stream tube. In this section we briefly discuss the highlights of results for variations in the bottom slope and forcing.

5.1. Bottom Slope

The presence of the bottom slope provides two important effects; it introduces both a background potential vorticity gradient, which allows isolated eddies to translate with shallow water on their right in the northern hemisphere, and a gravitational acceleration component along the bottom directed offshore. We have repeated the calculation described in section 4 using different bottom slopes in order to determine the importance of these two effects.

Figure 14 shows a sequence of the bottom and surface density anomaly fields over a flat bottom ($s = 0$). All other parameters are the same as in the standard case described in section 4. The development is qualitatively identical to the standard case. A density front forms near the edge of the forcing region. The front becomes unstable, initially creating a single eddy at the leading edge as in Figure 3, and then forming a number of eddies. However, there are two striking differences between the flat bottom case and the sloping bottom case (Figure 3). First, the eddies over the flat bottom tend to move in the $-x$ direction. Second, the eddies move offshore at a much slower rate than they did over the sloping bottom. As a consequence, the water remains beneath the forcing region for a longer time and becomes much denser before the offshore eddy transport becomes important in limiting the maximum density anomaly (Figure 15), yielding a maximum baroclinic Rossby radius of 6 km compared with 3.6 km in

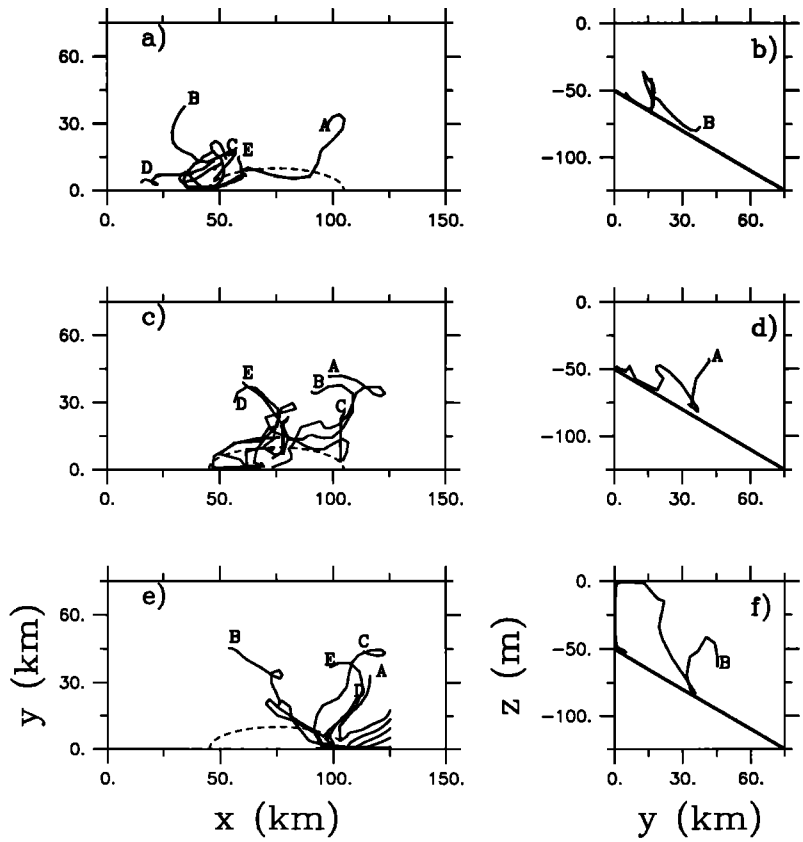

Figure 13. Same as in Figure 12, but for particles released 3 m above the bottom.

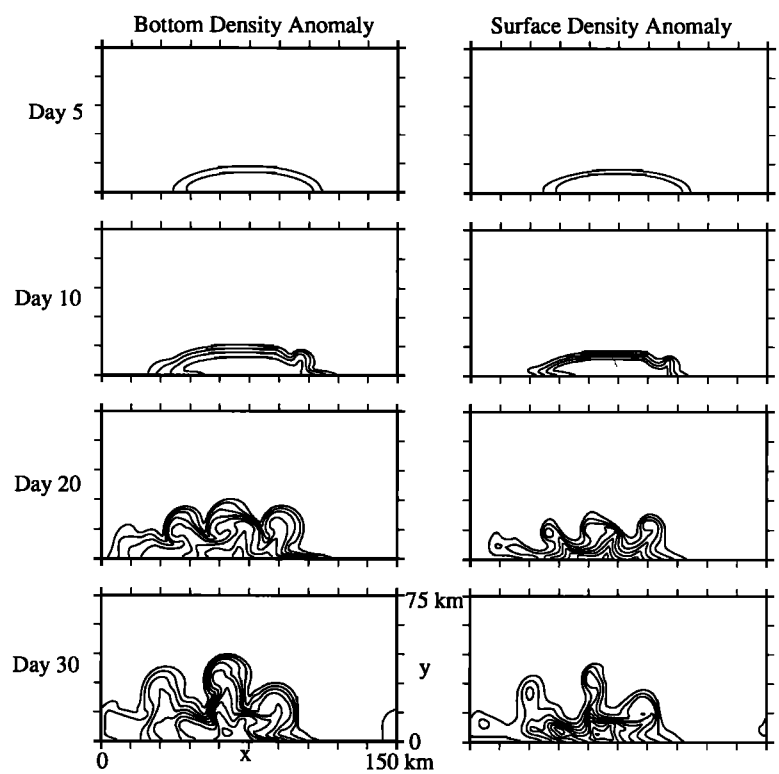

Figure 14. Plan views of the density anomaly at the (left) bottom and (right) surface at days 5, 10, 20, and 30 for the case with a flat bottom ($s = 0$). The contour interval is 0.1 kg m^{-3}, the most offshore contour being 0.1 kg m^{-3}.

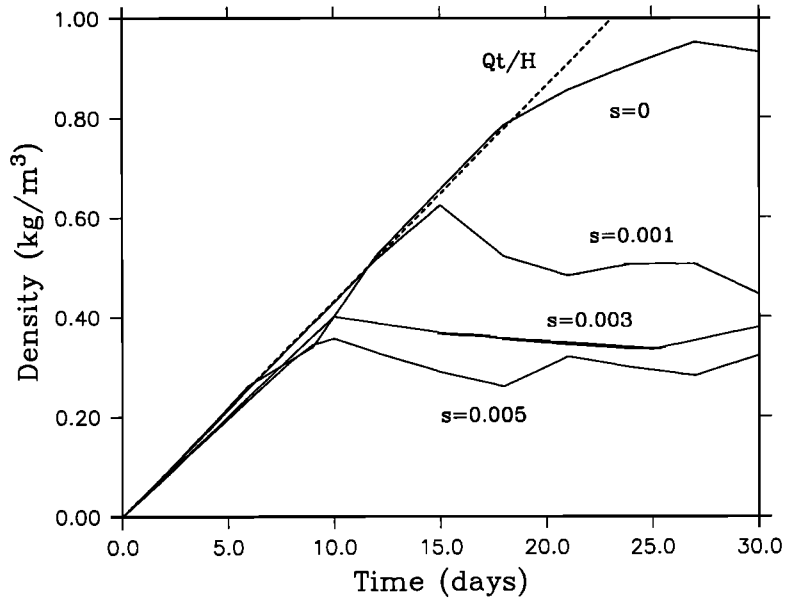

Figure 15. Time series of the maximum density anomaly within the computational domain for several bottom slopes (s = 0, 0.001, 0.003, and 0.005) and the standard choice of other parameters.

the sloping bottom case. This may seem counter intuitive because one typically expects the bottom slope to inhibit cross-isobath motion because it introduces an ambient potential vorticity gradient. Apparently, the additional gravitational acceleration component combined with the eddy dynamics dominate the offshore eddy movement. Steeper bottom slopes of $s = 0.003$ and $s = 0.005$ lead to even more rapid offshore movement of eddies, thereby shortening the time to instability and further reducing the maximum density anomaly (Figure 15).

For the steepest bottom slope ($s = 0.005$), eddies again develop near the leading edge of the forcing region, but they now travel even more rapidly offshore (Figure 16), thereby limiting the maximum density anomaly to 0.322 kg m^{-3}. Between days 10 and 20 the eddy of dense fluid which forms at the leading edge of the forcing region moves from roughly $x = 112$ km, $y = 23$ km to roughly $x = 131$ km, $y = 47$ km, yielding mean velocities of 0.028 m s^{-1} in the offshore direction and 0.022 m s^{-1} in the $+x$ direction. The maximum density within this eddy is 0.15 kg m^{-3} compared with 0.225 kg m^{-3} for the standard case. In addition, the flow becomes highly bottom trapped, as indicated by the vast difference between the surface and bottom density fields. The surface density anomaly after 30 days hardly extends outside the forcing region, whereas the bottom density anomaly has reached the offshore boundary. Figure 17 shows the highly bottom-trapped nature of the density field after 20 days. Because of this strong bottom trapping, the total offshore mass flux associated with the eddies is no greater than in the case of weaker bottom slopes, despite the dense water traveling more rapidly offshore. Figure 18 compares the mass fluxes across $y = 35$ km as a function of time for various bottom slopes. The fluxes exhibit no simple trend with increasing bottom slope, although the onset of the fluxes occurs sooner with increasing bottom slope. The flat bottom case clearly exhibits the smallest offshore mass flux, but the mass fluxes for bottom slopes of 0.001 and 0.005 are comparable, while the flux for a bottom slope of 0.003 is somewhat less. These differences probably represent a subtle interplay between the eddy dynamics and the downslope gravitational component for different bottom slopes.

5.2. Forcing Amplitude

To examine the influence of a decrease in forcing amplitude, the standard case in section 4 was repeated with $Q = 0.5 \times 10^{-5}$ kg m^{-2} s^{-1}, which is 1/5 of the standard value. The instabilities developed more slowly at the leading edge of the forcing region, becoming apparent at day 13. The time required for the initial instability to appear is roughly proportional to $Q^{-1/2}$. The maximum density increases as Qt/H for the entire 30-day time period. However, the behavior is qualitatively identical, with eddies again dominating the offshore transport, although the dense water is not carried as far offshore within 30 days. While it is possible that a qualitative change in behavior occurs for even weaker forcing, in practice, the timescales for the flow would be so long that processes neglected here, such as wind forcing, probably become important.

5.3. Length of Forcing Region

Two cases were examined in which the along-shelf length of the forcing region was altered; it was reduced to $a = 20$ km (with a 5-km half-cosine falloff at each end) and increased to $a = 100$ km (with a 25-km half-cosine falloff at each end). In both cases the cross-shelf scale of the forcing region was unchanged ($b = 10$ km with a half-cosine falloff of 5 km). In the first case the length scale of the instability ($\approx$15 km) was nearly the

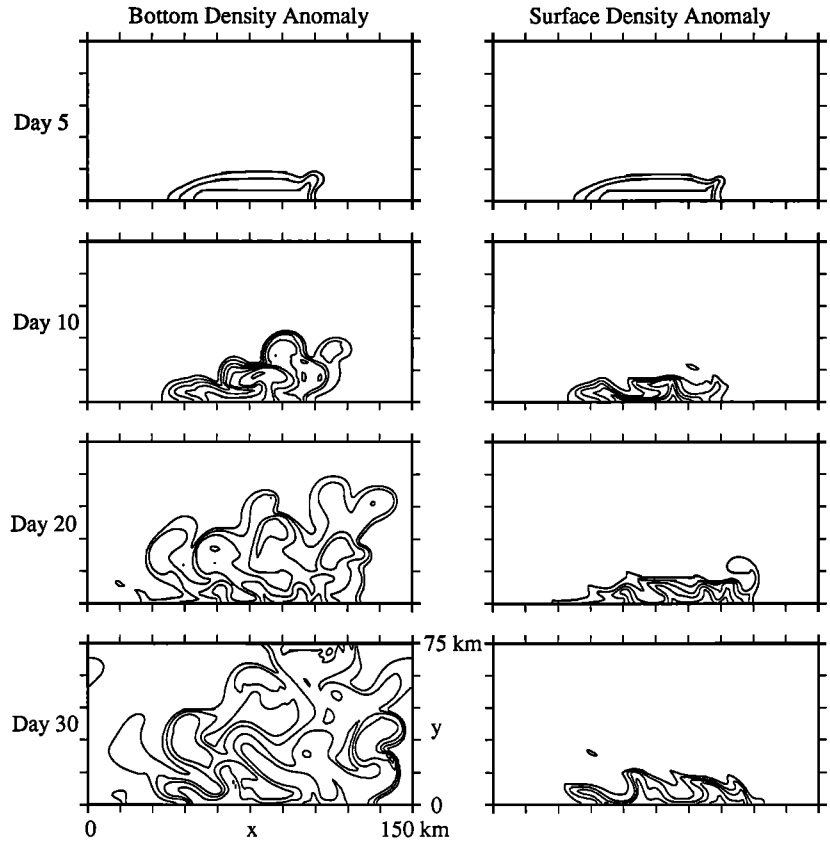

Figure 16. Plan views of the density anomaly at the (left) bottom and (right) surface at days 5, 10, 20, and 30 for the case with a bottom slope of $s = 0.005$. The contour interval is 0.05 kg m^{-3}, the most offshore contour being 0.05 kg m^{-3}.

size of the forcing region, so only one large eddy formed within the forcing region. As it slowly moved away, another eddy formed in the forcing region. In the latter case a series of eddies developed along the edge of the forcing region with horizontal scales similar to those in section 4. Thus the dominance of the eddy transport of the dense fluid is not affected by the length of the forcing region, although the number of eddies formed is apparently proportional to the along-shelf scale of the forcing region.

5.4. Other Parameters

Several model runs were performed with a bottom slope of 0.001 and the depth at the coastal boundary changed to 20 m. The density within the forcing region increased more rapidly, as one would expect, and the eddies developed more rapidly. Otherwise, the results were qualitatively identical to those of section 4. Several other model runs used constant values of the vertical mixing coefficient in the range of 0.001 to 0.005 m^2 s^{-1} without substantial changes in the results.

6. Discussion

The primary conclusion of this work is that eddy processes dominate the offshore transport of dense water from the coastal polynya region when a constant negative buoyancy forcing is applied over a limited along-

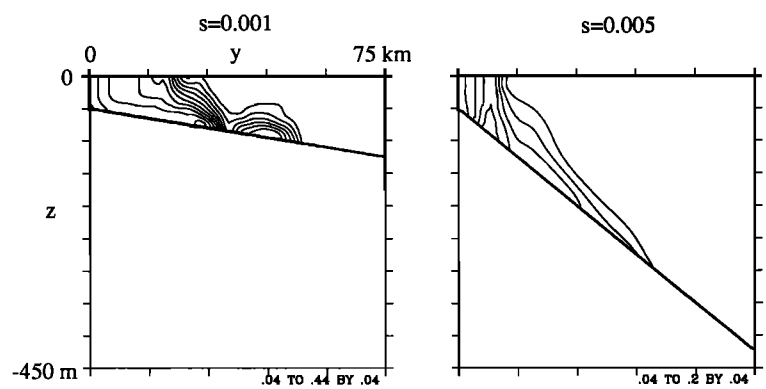

Figure 17. Cross-shelf sections of the density anomaly at $x = 75$ km on day 20 for the (left) standard case ($s = 0.001$) and (right) with the steepest bottom slope ($s = 0.005$).

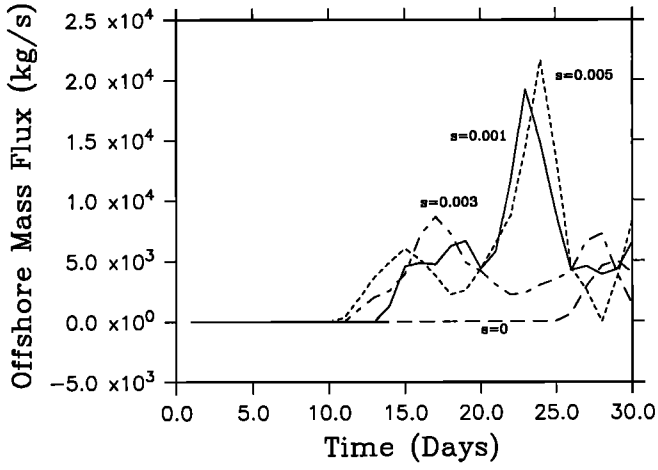

Figure 18. Time series of the offshore mass fluxes, computed from equation (24), across $y = 35$ km for various bottom slopes.

shelf extent. This is true for various bottom slopes, negative buoyancy forcing rates, and along-shelf scales for the forcing applicable to Arctic shelves. We found no evidence of a quasi-steady, stream-tube flow.

One consequence is that the spatial correlation scales for the dense water are small, of the order of 15 km or less. The horizontal scales are consistent with linear baroclinic instability theory (i.e., roughly 5 times the baroclinic Rossby radius) and are consistent with observations of shelf-derived eddies present in the central basins [*Newton et al.*, 1974; *Manley and Hunkins*, 1985]. The small spatial scales and complex dynamics associated with intermittent eddies make resolution of the flow field on an observational basis rather difficult. Furthermore, simultaneous direct measurements of the density and velocity fields are necessary because the flow near the bottom is ageostrophic owing to bottom friction (Figure 6), so geostrophic calculations from the density field would be misleading.

Another consequence of the domination of eddy processes is the large temporal variability in the offshore mass flux (Figures 11 and 18). The fluxes across an isobath vary drastically over short timescales as various eddies cross the isobath. This would make flux calculations from single hydrographic transects extremely misleading because they may vary by up to an order of magnitude over timescales of 1 to 3 days. Furthermore, the fluxes have horizontal scales in the along-shelf direction which are comparable to the eddy scales of 15 km, so high resolution sampling would be necessary to resolve the eddy contributions to the cross-shelf flux.

An important implication of the present results is that the offshore eddy transport of dense water severely limits the maximum attainable density anomaly within the polynya, despite continued negative buoyancy forcing. The maximum density anomaly increases linearly with time as Qt/H, only until the eddies (generated by frontal instabilities) move away from the forcing region (of the order of 5 to 10 days), after which the maximum density is roughly constant (Figure 15). This suggests that the maximum density of dense shelf water in the Arctic may not be determined by the strength of the forcing (i.e., by the ice production rate), but rather by the dynamics of the front formed at the edge of the polynya and the interaction with the local bottom topography. For the range of forcing, coastal depth, and bottom slopes used in this study the maximum density anomalies ranged from 0.2 to 1.0 kg m^{-3}. For the standard case with a gently sloping bottom ($s = 0.001$) the maximum density anomaly was roughly 0.45 kg m^{-3}, consistent with observations from St. Lawrence Island [*Schumacher et al.*, 1982].

There are numerous features of actual polynyas which have been neglected in this study in order to examine particular processes. For example, we have neglected ice effects, which produce frictional effects at the surface and set the cross-shelf scale for the brine rejection effects. The presence of surface stresses owing to ice would likely lead to less energetic velocities near the surface, as well as a more complicated secondary circulation pattern near the edge of the forcing region. It is possible that surface frictional processes might affect the sensitivity of the cross-channel fluxes to bottom slope by decoupling the eddies from the surface, perhaps leading to a monotonic increase in cross-isobath mass fluxes with increasing bottom slope.

We have also neglected wind forcing. Many coastal polynyas are opened up by the passage of synoptic scale weather systems and the offshore advection of ice [*Pease*, 1987] which would induce both additional boundary layer effects within the forcing region, as well as the generation of along-isobath motions. The former would affect the secondary circulation within the forcing region, while the latter could advect dense water out of the forcing region along isobaths, potentially reducing or even reversing some of the boundary currents described here.

The idealized nature of the forcing used here simplified the problem; however, the buoyancy forcing within polynyas is obviously a complex spatial and temporal pattern. Unfortunately, there are few observations to guide us in specifying a more realistic forcing, particularly on the small scales of variability (5-10 km) of the Arctic shelves. There is a clear need for measurements of the spatial and temporal scales of brine rejection rates within polynyas.

As mentioned above, we have neglected the effects of variation in along-shelf bathymetry. The presence of canyons, sills, and other topographic features may act to channel flows of dense water (e.g., *Quadfasel et al.*, 1988]. Future work will address this issue.

The processes of both lateral and vertical mixing, as well as shallow convection, are poorly understood in the coastal ocean, and the parameterizations used here are primarily for convenience, rather than an accurate representation of any observations. However, the spatial scales of the brine rejection and hence convection probably play an important role in local vertical mixing, and particle paths are likely to be much more dominated by turbulence than our idealized flows would indicate.

Tidal mixing processes, which were neglected here, are also likely to play a role in many regions of the Arctic.

This work suggests that coastal polynyas are potentially an important area for the formation of eddies on the Arctic shelves, and the spatial and velocity scales of the eddies in the model are consistent with those of shelf-derived water observed in the central basins, but much more work is necessary to understand how these eddies are transported to and across the shelf break in the presence of ambient stratification and steep topography. The interaction of the shelf eddies with the Arctic halocline is an important issue to address in the future.

Finally, one important technical point involving horizontal resolution of the model and cross-isobath transport is worth mentioning. During some preliminary model runs with a large along-shelf scale (100 km) in the forcing region a grid with a 4-km by 4-km resolution was used because of the length of the channel (400 km). The instabilities described above did not develop over a 30-day time period. When this run was repeated with a 2-km by 2-km grid spacing, eddies developed on a timescale similar to that in section 4. Further reduction in grid spacing did not alter the results substantially. The point is that horizontal resolution of the baroclinic radius of deformation is absolutely critical in resolving cross-isobath transport of dense fluid. The use of grids which do not resolve these eddies will not give reliable estimates for the cross-isobath mass fluxes. It will be important in the future to develop parameterizations of lateral exchange on continental shelves in the Arctic because it is unlikely that basin scale models with 1-km horizontal grid resolution will be practical soon.

7. Summary

A primitive equation numerical model (SPEM) has been used to examine the circulation and transport of dense water formed on a shallow, sloping continental shelf. Negative buoyancy forcing is imposed as a constant surface flux over a half-elliptical region adjacent to a coastal boundary with a scale of 60 km in the along-shelf direction. The flow field response essentially consists of three phases as follows: (1) geostrophic adjustment, during which the water within the forcing region becomes denser and the flow at the edge of the forcing region accelerates; (2) instability, during which the density front deforms at the leading edge of the forcing region (relative to Kelvin wave propagation) and eddies develop; and (3) offshore eddy transport, during which the dense fluid is transported offshore across isobaths. In all cases of bottom slopes and forcings, eddy transport was the dominant process in the transport of dense water offshore.

The eddies form in the following way. Bottom friction breaks the antisymmetry of the near-surface and near-bottom flow during the geostrophic adjustment process. This sets up horizontally convergent boundary currents at the leading edge of the forcing region which sharpen and deform the density front. Instabilities then form along the front and produce a complex eddy field. The offshore flux of dense water seaward of the forcing region begins sooner over a steeper bottom slope, but the total mass flux exhibits a complicated dependence on the bottom slope, with fluxes comparable for bottom slopes of 0.001 and 0.005, smaller for a bottom slope of 0.003, and smallest of all for a flat bottom.

Appendix: Resolution of the Bottom Boundary Layer

Given the importance of bottom friction in the present calculations, it is necessary to resolve the bottom boundary layer adequately. The choice of Chebyshev polynomials as a basis set for the vertical structure of SPEM is designed to enhance the ability of the model to resolve both surface and bottom boundary layers. However, the number of modes used in the calculations is arbitrary, so some objective means of assessing the performance is essential. Here we provide one way of evaluating the resolution of the surface or bottom boundary layers for a given number of Chebyshev modes.

The resolution of the bottom boundary layer may be examined by asking how well the Chebyshev modes represent the structure of an Ekman layer which develops beneath a uniform flow U subject to bottom boundary conditions (9) and (10). The velocity normal to U (and scaled by U) is

$$v_E/U = \exp[-(z+1)/\delta] \{B_1 \cos[(z+1)/\delta] + B_2 \sin[(z+1)/\delta]\} \tag{25}$$

where δ is the e-folding scale of the Ekman layer measured as a fraction of the total water depth. The bottom

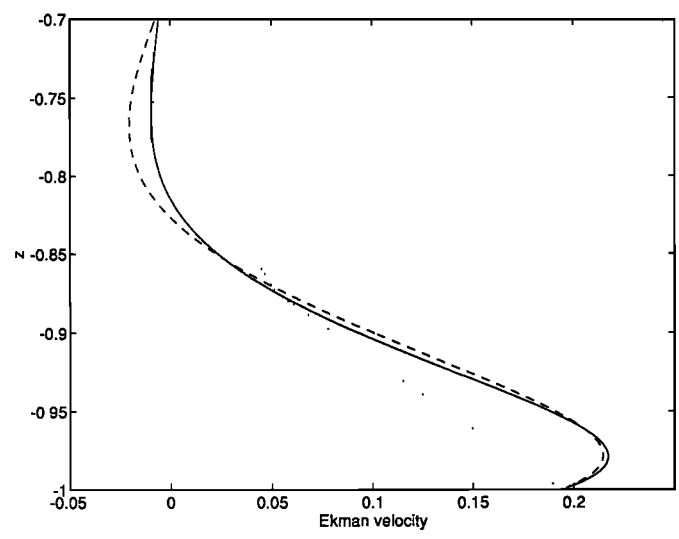

Figure A1. The vertical structure of an Ekman layer as a function of nondimensional depth (with $z = -1$ being the bottom) for the analytical solution (solid curve), a spectral expansion in Chebyshev polynomials with nine modes (dashed curve), and an evenly spaced grid representation with $\Delta z = 0.125$ (dotted curve). The Ekman decay scale is $\delta = 0.07$.

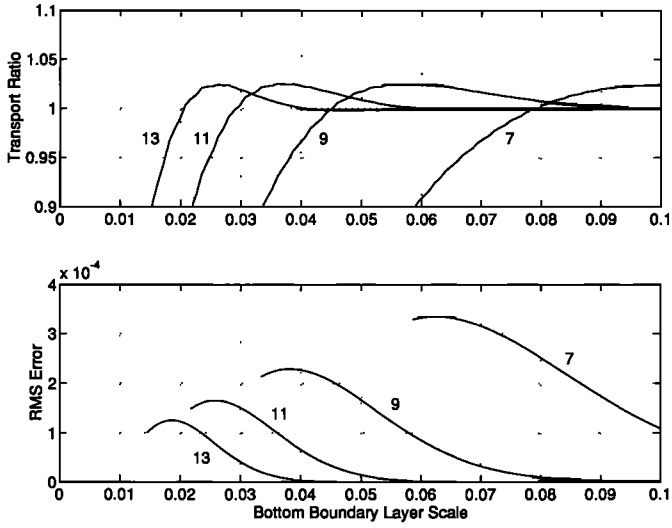

Figure A2. The error in representing an Ekman layer as a function of the bottom boundary layer scale δ and the number of Chebyshev modes used in the expansion $(N+1)$. The error is measured by (top) the ratio of the Ekman transport (over 1.5 δ) in the Chebyshev expansion to the analytical value and (bottom) the rms difference over the entire water column.

is at $z=-1$. The coefficients $B_{1,2}$ are given by

$$B_1 = \frac{\beta}{(1+\beta)^2+\beta^2}; \qquad B_2 = \frac{1+\beta}{(1+\beta)^2+\beta^2} \quad (26)$$

where $\beta = (A_V f/2r^2)^{1/2}$ is a measure of the relative importance of vertical mixing and bottom friction in (9) and (10). The classical Ekman layer is recovered when $\beta = 0$. If we assume that we know v_E/U at the collocation points (i.e., the extrema of the Chebyshev polynomials), then v_E/U can be estimated using $N+1$ Chebyshev modes as done in SPEM (see *Hedstrom* [1990] for details). Figure A1 shows a comparison using nine Chebyshev modes (i.e., $N=8$) for $\delta = 0.07$ and $\beta = 1.14$, the values used in the standard case in section 4. Also shown in Figure A1 is the approximation of v_E/U based on nine equally spaced grid points. The choice of $\delta = 0.07$ means that the boundary layer e-folding scale is 7% of the water column and is obtained using the water depth from the deepest part of the model domain ($h = 125$ m) for the standard case in section 4. Clearly, the Chebyshev modes resolve the Ekman layer solution quite well, whereas the grid points do not. The Chebyshev modes, being continuous functions, contain information between the grid points. So nine Chebyshev modes easily resolve the bottom boundary layer in the standard calculation described in section 4.

The resolution issue can be generalized somewhat by defining appropriate measures of error and computing them for a variety of values of δ and N. We have chosen the rms difference between v_E/U and the Chebyshev approximation as one measure. A second error measure is based on the transport in the Ekman layer (i.e., the vertical integral of v_E/U). About 90% of the transport in the Ekman layer occurs within 1.5 e-folding scales from the boundary. Therefore we compute the integral of v_E/U from $z=-1$ to $z=-1+1.5\delta$ and compare it with the estimate of the integral based on the Chebyshev representation. The ratio of the two estimates is a measure of the error in the Ekman transport introduced by the modal expansion (i.e., a ratio of 1 would mean perfect agreement). The results are shown in Figure A2 as a function of δ for several choices of N. There is no obvious, objective cutoff for adequate resolution, but visual comparisons (as in Figure A1) suggest that when the transport ratio drops below 1, the resolution is inadequate. This corresponds roughly to an rms error of less than 2×10^{-4}. So Figure A2 could be used to estimate the number of Chebyshev modes needed to resolve a particular boundary boundary layer in a given total water depth. For example, resolving a 10-m bottom boundary layer in 200 m of water ($\delta = 0.02$) would require at least 13 modes. Of course, the error estimates in Figure A2 would change for a different choice of β (i.e., A_V, f, or r), but the sensitivity to β is fairly weak.

Acknowledgments. We thank Breck Owens for initial encouragement on this problem and Jim Price, Larry Pratt, and Dale Haidvogel for helpful discussions about various aspects of this work. We also appreciate the comments of the two anonymous reviewers. This work was supported by the National Science Foundation as part of the Arctic Systems Science (ARCSS) Initiative, under grant DPP-9113940. We performed most of the computations at the National Center for Atmospheric Research in Boulder, Colorado, which is funded by the National Science Foundation. Woods Hole Oceanographic Institution contribution 8588.

References

Aagaard, K., L. K. Coachman, and E. Carmack, On the halocline of the Arctic Ocean, *Deep Sea Res., Part A, 28A*, 529–545, 1981.

Aagaard, K., J. H. Swift, and E. C. Carmack, Thermohaline circulation in the Arctic Mediterranean Seas, *J. Geophys. Res., 90*, 4833–4846, 1985.

Blindheim, J., Cascading of Barents Sea bottom water into the Norwegian Sea, *Rapp. P.V. Reun. Cons. Int. Explor. Mer., 188*, 49–58, 1989.

Dewar, W. K., and P. D. Killworth, On the cylinder collapse problem, mixing, and the merger of isolated eddies, *J. Phys. Oceanogr., 20*, 1563–1575, 1990.

Duffy, D. G., The application of the semigeostrophic equations to the frontal instability problem, *J. Atmos. Sci., 33*, 2322–2337, 1976.

Haidvogel, D., J. Wilkin, and R. Young, A semi-spectral primitive equation ocean circulation model using vertical sigma and orthogonal curvilinear horizontal coordinates, *J. Comput. Phys., 94*, 151–185, 1991.

Hedstrom, K., User's manual for a semi-spectral primitive equation regional ocean-circulation model version 3.0 B, *Tech. Rep. SR-1*, Inst. for Naval Oceanogr., 82 pp., Stennis Space Ctr., Miss., 1990.

Hogg, N. G., and H. M. Stommel, The heton, an elementary interaction between discrete baroclinic geostrophic vor-

tices, and its implications concerning eddy heat-flow, *Proc. R. Soc. London A, 397*, 1–20, 1985.

James, I. D., A model of the annual cycle of temperature in a frontal region of the Celtic Sea, *Estuarine Coastal Mar. Sci., 5*, 339–353, 1977.

Jones, H., and J. Marshall, Convection with rotation in a neutral ocean: A study of open-ocean deep convection, *J. Phys. Oceanogr., 23*, 1009–1039, 1993.

Kowalik, Z., and J. B. Matthews, Numerical study of the water movement driven by brine rejection from nearshore Arctic ice, *J. Geophys. Res., 88*, 2953–2958, 1983.

Manley, T. O., and K. Hunkins, Mesoscale eddies of the Arctic Ocean, *J. Geophys. Res., 90*, 4911–4930, 1985.

Martin, S., and D. J. Cavalieri, Contributions of the Siberian shelf polynyas to the Arctic Ocean intermediate and deep water, *J. Geophys. Res., 94*, 12,725–12,738, 1989.

Melling, H., The formation of a haline shelf front in wintertime in an ice covered Arctic Sea, *Cont. Shelf Res., 13*, 1123–1147, 1993.

Melling, H., and E. L. Lewis, Shelf drainage flows in the Beaufort Sea and their effect on the Arctic Ocean pycnocline, *Deep Sea Res., Part A, 29*, 967–985, 1982.

Midttun, L., Formation of dense bottom water in the Barents Sea, *Deep Sea Res., Part A, 32*, 1233–1241, 1985.

Muench, R. D., J. D. Schumacher, and S. A. Salo, Winter currents and hydrographic conditions on the northern central Bering Sea shelf, *J. Geophys. Res., 93*, 516–526, 1988.

Munk, W. H., and E. H. Anderson, Notes on a theory of the thermocline, *J. Marine Res., 7*, 276–295, 1948.

Nansen, F., Northern waters: Captain Roald Amundsen's oceanographic observations in the Arctic Seas in 1901, *Skr. Nor. Vidensk. Akad. Kl. 1: Mat.-Naturvidensk, Kl., 3*, 145 pp., 1906.

Newton, J. L., K. Aagaard, and L. K. Coachman, Baroclinic eddies in the Arctic Ocean, *Deep Sea Res., 21*, 707–719, 1974.

Nof, D., The translation of isolated cold eddies on a sloping bottom, *Deep Sea Res., Part A*, 171–182, 1983.

Nof, D., and L. M. Simon, Laboratory experiments on the merging of anticyclonic eddies, *J. Phys. Oceanogr., 17*, 343–357, 1987.

Pease, C. H., The size of wind-driven coastal polynyas, *J. Geophys. Res., 92*, 7049–7059, 1987.

Pedlosky, J., *Geophysical Fluid Dynamics*, 2nd ed., 703 pp., Springer-Verlag, New York, 1987.

Quadfasel, D., B. Rudels, and K. Kurz, Outflow of dense water from a Svalbard fjord into the Fram Strait, *Deep Sea Res., Part A, 35*, 1143–1150, 1988.

Schumacher, J. D., K. Aagaard, C. H. Pease, and R. B. Tripp, Effects of a shelf polynya on flow and water properties in the northern Bering Sea, *J. Geophys. Res., 88*, 2723–2732, 1983.

Smith, P. C., A stream-tube model for bottom boundary currents in the ocean, *Deep Sea Res., 22*, 853–873, 1975.

Swift, J. H., T. Takahashi, and H. D. Livingston, The contribution of the Greenland and Barents Seas to the deep water of the Arctic Ocean, *J. Geophys. Res., 88*, 5981–5986, 1983.

Zabusky, N. J., and J. C. McWilliams, A modulated point-vortex model for geostrophic, beta-plane dynamics, *Phys. Fluids, 25*, 2175–2182, 1982.

D. C. Chapman and G. G. Gawarkiewicz, Physical Oceanography Department, Woods Hole Oceanographic Institution, Woods Hole, MA 02543

(Received October 29, 1993; revised June 17, 1994; accepted June 29, 1994.)